

iBGP and eBGP

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Purpose

In this lab, I explored and learned more about BGP by researching iBGP, such as its purpose and how to set it up on a network. I learned to do this in both IPv4 and IPv6 and implemented these networks with loopback addresses in GNS3.

Background Information

General Overview

Briefly, iBGP is BGP used as an internal mechanism to exchange BGP information amongst multiple BGP border routers. If there is a BGP session between two BGP peers in the same AS, they are an iBGP session. eBGP is the standard form of BGP. It is used to route between routers that belong to different ASs.

iBGP Full Mesh Requirement

Unlike BGP that uses AS_PATH for loop detection and prevention (since the Autonomous System Numbers are added when there is an advertisement to an eBGP neighbor), iBGP peers do not need to do this because the Network Layer Reachability Information would not pass the validity check and would not install the prefix into the routing table. There are no other methods in existence that are able to detect loops with iBGP sessions.

Peering with Loopback Addresses

By default, BGP sessions are sourced by outbound interfaces to the IP addresses of BGP peers. Sometimes there is a loss of connectivity because iBGP doesn't advertise learned routes from other peers. One of the solutions to solve this is configuring loopback interfaces and BGP neighbors in order to create a session to the remote router's loopback address. This is the more efficient method to use in order to solve this link failure issue. With loopback set up for iBGP, the session will stay intact if there is ever a link failure because loopbacks are virtual interfaces that always stay on. If you do not statically set the source of BGP packets to an interface's primary IP address with the command "*neighbor ip-address update-source interface-type interface-number*," you may still have packets that are sourced from outbound interfaces and thus, are not able to correlate to the source IP addresses found in the BGP neighbor table.

Lab Summary

With this lab, I set up five Cisco 3600 routers with IPv4 and IPv6 addresses. Of the five routers, two were configured in eBGP, one in iBGP, and one with both eBGP and iBGP who worked as border routers between the different BGP areas. All routers that had iBGP also had OSPF set up between them. I configured a loopback on each router in interface Lo0 using the networks 1.0.0.0/32, 2.0.0.0/32, 3.0.0.0/32, 4.0.0.0/32, and 5.0.0.0/32 for their IPv4 addresses and 2001:acad:a:: to 2001:acad:e:: for their IPv6 addresses. For the interfaces between each router, I used the F0/1 and F0/0 and made the networks 10.0.0.0/24, 11.0.0.0/24, 12.0.0.0/24, and 13.0.0.0/24 for IPv4 and 2001:a::, 2001:b::, 2001:c::, and 2001:d:: for IPv6.

Lab Commands

Address-Family IPv6: Puts your router into address family configuration mode when inside a routing protocol session like BGP

Neighbor *IP Address* Activate: Used under an address family and allows BGP neighbors and peers to exchange information

Neighbor *IP Address* eBGP-Multihop #: Establish BGP connections to external peers in the network that are not directly connected to the device you are on

Neighbor *IP Address* Remote-AS #: Adds a device to the BGP or multiprotocol BGP neighbor table

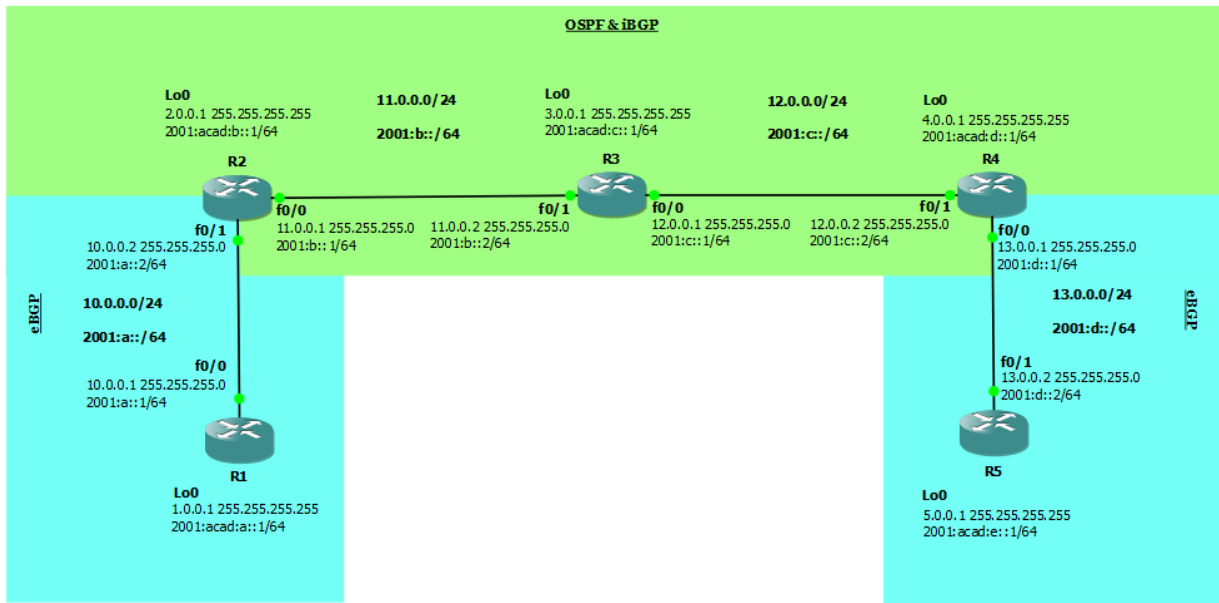
Neighbor *IP Address* Update-Source *Interface*: Allows the BGP sessions to use any working interfaces for TCP connections

Redistribute Static: Advertises routes known through a routing protocol to another routing protocol

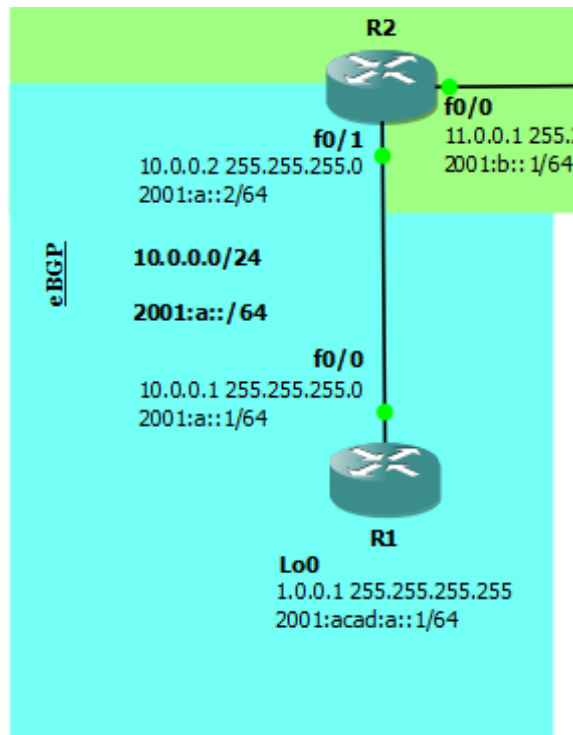
Show IP BGP: Displays entries in the BGP routing table

Show IP BGP Summary: Displays the status of all BGP connections

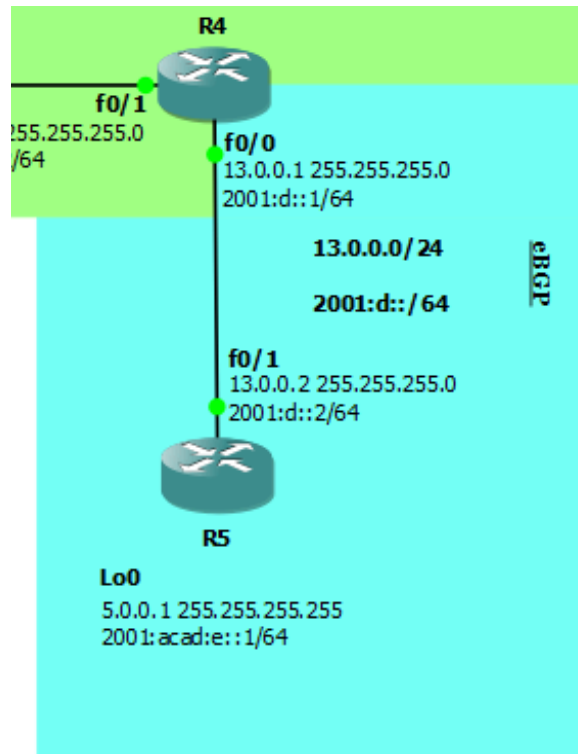
Network Diagram



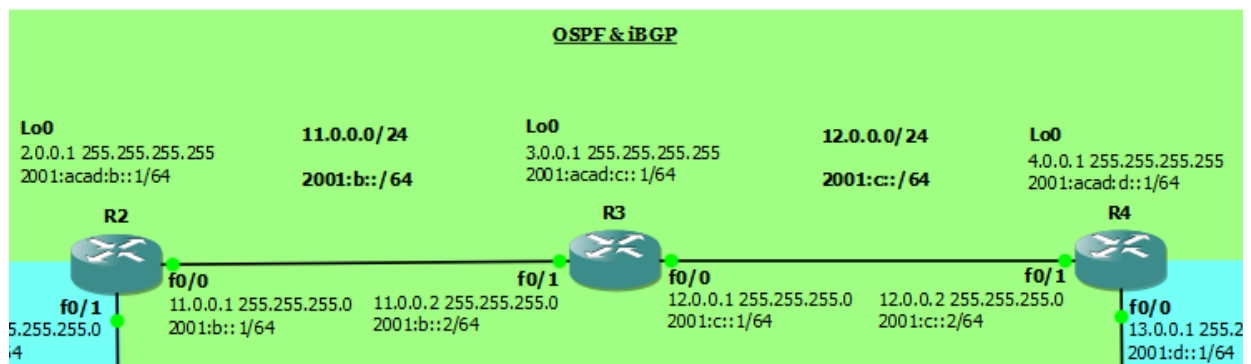
eBGP (Left Side)



eBGP (Right Side)



OSPF & iBGP



Configurations

Router 1:

- **Show Run**

```
ipv6 unicast-routing

interface Loopback0
 ip address 1.0.0.1 255.255.255.255
 ipv6 address 2001:ACAD:A::1/64

interface FastEthernet0/0
 ip address 10.0.0.1 255.255.255.0
 ipv6 address 2001:A::1/64

router bgp 10
 no synchronization
 bgp router-id 1.1.1.1
 bgp log-neighbor-changes
 network 1.0.0.0 mask 255.255.255.255
 network 10.0.0.0 mask 255.255.255.0
 neighbor 2.0.0.1 remote-as 20
 neighbor 2.0.0.1 ebgp-multihop 2
 neighbor 2.0.0.1 update-source Loopback0
 neighbor 2001:ACAD:B::1 remote-as 20
 neighbor 2001:ACAD:B::1 ebgp-multihop 2
 neighbor 2001:ACAD:B::1 update-source Loopback0
 no auto-summary

address-family ipv6
 neighbor 2001:ACAD:B::1 activate
 network 2001:A::/64
 network 2001:ACAD:A::/64
 exit-address-family

ip route 2.0.0.1 255.255.255.255 10.0.0.2

ipv6 route 2001:ACAD:B::/64 2001:A::2
```

- **Show IP BGP**

Network	Next Hop	Metric	LocPrf
Weight Path			
r 1.0.0.1/32	32.1.172.173	0	
0 20 ?			
r>	2.0.0.1	0	
0 20 ?			
* 3.0.0.1/32	32.1.172.173	0	
0 20 ?			
*>	2.0.0.1	0	
0 20 ?			
* 4.0.0.1/32	32.1.172.173	0	
0 20 ?			
*>	2.0.0.1	0	
0 20 ?			
* 5.0.0.1/32	32.1.172.173		
0 20 ?			
*>	2.0.0.1		
0 20 ?			
* 10.0.0.0/24	32.1.172.173	0	
0 20 i			
*	2.0.0.1	0	
0 20 i			
*>	0.0.0.0	0	
32768 i			
* 11.0.0.0/24	32.1.172.173	0	
0 20 i			
*>	2.0.0.1	0	
0 20 i			
* 12.0.0.0/24	32.1.172.173	2	
0 20 ?			
*>	2.0.0.1	2	
0 20 ?			
* 13.0.0.0/24	32.1.172.173		
0 20 i			
*>	2.0.0.1		
0 20 i			

- **Show IP BGP Summary**

BGP router identifier 1.1.1.1, local AS number 10


```

BGP table version is 13, main routing table version 13
8 network entries using 936 bytes of memory
17 path entries using 884 bytes of memory
7/6 BGP path/bestpath attribute entries using 868 bytes of
memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2712 total bytes of memory
BGP activity 20/3 prefixes, 32/4 paths, scan interval 60
secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
Up/Down	State/PfxRcd						
2.0.0.1	4	20	9861	9854	13	0	0
6d20h	8						
2001:ACAD:B::1	4	20	9867	9856	13	0	0
6d20h	8						

- **Show IP Route**

```

Gateway of last resort is not set

```

```

      1.0.0.0/32 is subnetted, 1 subnets
C        1.0.0.1 is directly connected, Loopback0
      2.0.0.0/32 is subnetted, 1 subnets
S        2.0.0.1 [1/0] via 10.0.0.2
      3.0.0.0/32 is subnetted, 1 subnets
B        3.0.0.1 [20/0] via 2.0.0.1, 6d20h
      4.0.0.0/32 is subnetted, 1 subnets
B        4.0.0.1 [20/0] via 2.0.0.1, 6d20h
      5.0.0.0/32 is subnetted, 1 subnets
B        5.0.0.1 [20/0] via 2.0.0.1, 6d20h
      10.0.0.0/24 is subnetted, 1 subnets
C        10.0.0.0 is directly connected, FastEthernet0/0
      11.0.0.0/24 is subnetted, 1 subnets
B        11.0.0.0 [20/0] via 2.0.0.1, 6d20h
      12.0.0.0/24 is subnetted, 1 subnets
B        12.0.0.0 [20/2] via 2.0.0.1, 6d20h
      13.0.0.0/24 is subnetted, 1 subnets
B        13.0.0.0 [20/0] via 2.0.0.1, 6d20h

```

- **Show IPv6 Route**

```
C   2001:A::/64 [0/0]
    via ::, FastEthernet0/0
L   2001:A::1/128 [0/0]
    via ::, FastEthernet0/0
B   2001:B::/64 [20/0]
    via 2001:ACAD:B::1
B   2001:C::/64 [20/0]
    via 2001:ACAD:B::1
B   2001:D::/64 [20/0]
    via 2001:ACAD:B::1
C   2001:ACAD:A::/64 [0/0]
    via ::, Loopback0
L   2001:ACAD:A::1/128 [0/0]
    via ::, Loopback0
S   2001:ACAD:B::/64 [1/0]
    via 2001:A::2
B   2001:ACAD:C::/64 [20/0]
    via 2001:ACAD:B::1
B   2001:ACAD:D::/64 [20/0]
    via 2001:ACAD:B::1
B   2001:ACAD:E::/64 [20/0]
    via 2001:ACAD:B::1
L   FE80::/10 [0/0]
    via ::, Null0
L   FF00::/8 [0/0]
    via ::, Null0
```

- **Ping R2**

- **F0/0**

- **11.0.0.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.0.0.1,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/31/40 ms
```

- **2001:b::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 20/32/52 ms
```

- **F0/1**

- **10.0.0.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/36/56 ms
```

- **2001:a::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::2,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 0/27/44 ms
```

- **Lo0**

- **2.0.0.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.0.0.1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/26/36 ms
```

- **2001:acad:b::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:B::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/28/48 ms

- **Ping R3**

- **F0/0**

- **12.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/58/72 ms

- **2001:c::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 40/51/84 ms

- **F0/1**

- **11.0.0.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.0.0.2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/60/84 ms

- **2001:b::2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:B::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 40/49/60 ms

- **Lo0**

- **3.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/56/108 ms

- **2001:acad:c::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:C::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/55/76 ms

- **Ping R4**

- **F0/0**

- **13.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 13.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 64/133/220 ms

- **2001:d::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:D::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 80/86/92 ms

- **F0/1**

- **12.0.0.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 88/95/100 ms

- **2001:c::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 52/89/120 ms

- **Lo0**

- **4.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 40/80/104 ms

- **2001:acad:d::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:D::1,
timeout is 2 seconds:
!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 84/90/96 ms

- **Ping R5**

- **F0/1**

- **13.0.0.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 13.0.0.2,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 120/130/144 ms

- **2001:d::2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:D::2,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 100/120/140 ms

- **Lo0**

- **5.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.0.0.1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 96/128/164 ms

- **2001:acad:e::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:ACAD:E::1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 100/115/136 ms

Router 2:

- **Show Run**

```
ipv6 unicast-routing

interface Loopback0
 ip address 2.0.0.1 255.255.255.255
 ipv6 address 2001:ACAD:B::1/64

interface FastEthernet0/0
 ip address 11.0.0.1 255.255.255.0
 ipv6 address 2001:B::1/64
 ipv6 ospf 1 area 0

interface FastEthernet0/1
 ip address 10.0.0.2 255.255.255.0
 ipv6 address 2001:A::2/64

router ospf 10
 router-id 2.2.2.2
 log-adjacency-changes
 redistribute bgp 20 subnets
 network 11.0.0.0 0.0.0.255 area 0

router bgp 20
 no synchronization
 bgp router-id 2.2.2.2
 bgp log-neighbor-changes
 network 2.0.0.0 mask 255.255.255.255
 network 10.0.0.0 mask 255.255.255.0
 network 11.0.0.0 mask 255.255.255.0
 redistribute static
 redistribute ospf 10
 neighbor 1.0.0.1 remote-as 10
 neighbor 1.0.0.1 ebgp-multihop 2
 neighbor 1.0.0.1 update-source Loopback0
 neighbor 3.0.0.1 remote-as 20
 neighbor 3.0.0.1 update-source Loopback0
```



```

neighbor 4.0.0.1 remote-as 20
neighbor 4.0.0.1 update-source Loopback0
neighbor 2001:ACAD:A::1 remote-as 10
neighbor 2001:ACAD:A::1 ebgp-multihop 2
neighbor 2001:ACAD:A::1 update-source Loopback0
neighbor 2001:ACAD:C::1 remote-as 20
neighbor 2001:ACAD:C::1 update-source Loopback0
neighbor 2001:ACAD:D::1 remote-as 20
neighbor 2001:ACAD:D::1 update-source Loopback0
no auto-summary

```

```

address-family ipv6
  neighbor 2001:ACAD:A::1 activate
  neighbor 2001:ACAD:C::1 activate
  neighbor 2001:ACAD:D::1 activate
  network 2001:A::/64
  network 2001:B::/64
  network 2001:ACAD:B::/64
  redistribute static
  redistribute ospf 10
  no synchronization
exit-address-family

```

```

ip route 1.0.0.1 255.255.255.255 10.0.0.1
ip route 3.0.0.1 255.255.255.255 11.0.0.2
ip route 4.0.0.1 255.255.255.255 12.0.0.2

```

```

ipv6 route 2001:ACAD:A::/64 2001:A::1
ipv6 route 2001:ACAD:C::/64 2001:B::2
ipv6 route 2001:ACAD:D::/64 2001:C::2
ipv6 router ospf 1

```

- **Show IP BGP**

Network	Next Hop	Metric	LocPrf
Weight Path			
*> 1.0.0.1/32	10.0.0.1	0	
32768 ?			
* i3.0.0.1/32	12.0.0.1	0	100
0 ?			

* i	12.0.0.1	0	100
0 ?			
*>	11.0.0.2	0	
32768 ?			
*> 4.0.0.1/32	0.0.0.0	0	
32768 ?			
* i5.0.0.1/32	13.0.0.2	0	100
0 ?			
*>i	13.0.0.2	0	100
0 ?			
* 10.0.0.0/24	32.1.172.173	0	
0 10 i			
*	1.0.0.1	0	
0 10 i			
*>	0.0.0.0	0	
32768 i			
* i11.0.0.0/24	12.0.0.1	2	100
0 ?			
* i	12.0.0.1	2	100
0 ?			
* i	32.1.172.173	0	100
0 i			
* i	3.0.0.1	0	100
0 i			
*>	0.0.0.0	0	
32768 i			
* i12.0.0.0/24	32.1.172.173	0	100
0 i			
* i	4.0.0.1	0	100
0 i			
Network	Next Hop	Metric	LocPrf
Weight Path			
* i	32.1.172.173	0	100
0 i			
* i	3.0.0.1	0	100
0 i			
*>	11.0.0.2	2	
32768 ?			
r i13.0.0.0/24	32.1.172.173	0	100
0 i			

```

r>i                               4.0.0.1                               0    100
0 i

```

- **Show IP BGP Summary**

```

BGP router identifier 2.2.2.2, local AS number 20
BGP table version is 12, main routing table version 12
8 network entries using 936 bytes of memory
22 path entries using 1144 bytes of memory
8/5 BGP path/bestpath attribute entries using 992 bytes of
memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3096 total bytes of memory
BGP activity 21/4 prefixes, 45/6 paths, scan interval 60
secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
Up/Down	State/PfxRcd						
1.0.0.1	4	10	9859	9866	12	0	0
6d20h	1						
3.0.0.1	4	20	9859	9867	12	0	0
6d20h	2						
4.0.0.1	4	20	9866	9866	12	0	0
6d20h	5						
2001:ACAD:A::1	4	10	9860	9872	12	0	0
6d20h	1						
2001:ACAD:C::1	4	20	9860	9871	12	0	0
6d20h	2						
2001:ACAD:D::1	4	20	9870	9870	12	0	0
6d20h	5						

- **Show IP Route**

```

Gateway of last resort is not set

```

```

      1.0.0.0/32 is subnetted, 1 subnets
S      1.0.0.1 [1/0] via 10.0.0.1
      2.0.0.0/32 is subnetted, 1 subnets
C      2.0.0.1 is directly connected, Loopback0

```

```

    3.0.0.0/32 is subnetted, 1 subnets
S      3.0.0.1 [1/0] via 11.0.0.2
    4.0.0.0/32 is subnetted, 1 subnets
S      4.0.0.1 [1/0] via 12.0.0.2
    5.0.0.0/32 is subnetted, 1 subnets
B      5.0.0.1 [200/0] via 13.0.0.2, 6d20h
    10.0.0.0/24 is subnetted, 1 subnets
C      10.0.0.0 is directly connected, FastEthernet0/1
    11.0.0.0/24 is subnetted, 1 subnets
C      11.0.0.0 is directly connected, FastEthernet0/0
    12.0.0.0/24 is subnetted, 1 subnets
O      12.0.0.0 [110/2] via 11.0.0.2, 6d20h,
FastEthernet0/0
    13.0.0.0/24 is subnetted, 1 subnets
O E2    13.0.0.0 [110/1] via 11.0.0.2, 6d20h,
FastEthernet0/0

```

- **Show IPv6 Route**

```

C    2001:A::/64 [0/0]
    via ::, FastEthernet0/1
L    2001:A::2/128 [0/0]
    via ::, FastEthernet0/1
C    2001:B::/64 [0/0]
    via ::, FastEthernet0/0
L    2001:B::1/128 [0/0]
    via ::, FastEthernet0/0
O    2001:C::/64 [110/2]
    via FE80::CE03:40FF:FE64:1, FastEthernet0/0
B    2001:D::/64 [200/0]
    via 2001:ACAD:D::1
S    2001:ACAD:A::/64 [1/0]
    via 2001:A::1
C    2001:ACAD:B::/64 [0/0]
    via ::, Loopback0
L    2001:ACAD:B::1/128 [0/0]
    via ::, Loopback0
S    2001:ACAD:C::/64 [1/0]
    via 2001:B::2
S    2001:ACAD:D::/64 [1/0]
    via 2001:C::2

```

```
B    2001:ACAD:E::/64 [200/0]
      via 2001:D::2
L    FE80::/10 [0/0]
      via ::, Null0
L    FF00::/8 [0/0]
      via ::, Null0
```

- **Ping R1**

- **F0/0**

- **10.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/28/44 ms

- **2001:a::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/27/44 ms

- **Lo0**

- **1.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/28/32 ms

- **2001:acad:a::1**

Type escape sequence to abort.

```
Sending 5, 100-byte ICMP Echos to 2001:ACAD:A::1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 0/26/48 ms
```

- **Ping R3**

- **F0/0**

- **12.0.0.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/28/44 ms
```

- **2001:c::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/24/44 ms
```

- **F0/1**

- **11.0.0.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.0.0.2,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/29/40 ms
```

- **2001:b::2**

```
Type escape sequence to abort.
```

Sending 5, 100-byte ICMP Echos to 2001:B::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/23/32 ms

- **Lo0**

- **3.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/32/44 ms

- **2001:acad:c::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:C::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 20/27/36 ms

- **Ping R4**

- **F0/0**

- **13.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 13.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/60/76 ms

- **2001:d::1**

Type escape sequence to abort.

```
Sending 5, 100-byte ICMP Echos to 2001:D::1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/55/60 ms
```

- **F0/1**

- **12.0.0.2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.2,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 36/55/76 ms
```

- **2001:c::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/56/64 ms
```

- **Lo0**

- **4.0.0.1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.0.0.1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/59/76 ms
```

- **2001:acad:d::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:D::1,
timeout is 2 seconds:
!!!!!!
```


Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/55/88 ms

- **Ping R5**

- **F0/1**

- **13.0.0.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 13.0.0.2,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 64/88/96 ms

- **2001:d::2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:D::2,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 72/88/104 ms

- **Lo0**

- **5.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.0.0.1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 76/95/116 ms

- **2001:acad:e::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:ACAD:E::1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/76/96 ms

Router 3:

- **Show Run**

```
ipv6 unicast-routing

interface Loopback0
 ip address 3.0.0.1 255.255.255.255
 ipv6 address 2001:ACAD:C::1/64

interface FastEthernet0/0
 ip address 12.0.0.1 255.255.255.0
 ipv6 address 2001:C::1/64
 ipv6 ospf 1 area 0

interface FastEthernet0/1
 ip address 11.0.0.2 255.255.255.0
 ipv6 address 2001:B::2/64
 ipv6 ospf 1 area 0

router ospf 10
 router-id 3.3.3.3
 log-adjacency-changes
 network 11.0.0.0 0.0.0.255 area 0
 network 12.0.0.0 0.0.0.255 area 0

router bgp 20
 no synchronization
 bgp router-id 3.3.3.3
 bgp log-neighbor-changes
 network 3.0.0.0 mask 255.255.255.255
 network 11.0.0.0 mask 255.255.255.0
 network 12.0.0.0 mask 255.255.255.0
 neighbor 2.0.0.1 remote-as 20
 neighbor 2.0.0.1 update-source Loopback0
 neighbor 4.0.0.1 remote-as 20
 neighbor 4.0.0.1 update-source Loopback0
 neighbor 2001:ACAD:B::1 remote-as 20
 neighbor 2001:ACAD:B::1 update-source Loopback0
```

```
neighbor 2001:ACAD:D::1 remote-as 20
neighbor 2001:ACAD:D::1 update-source Loopback0
no auto-summary
```

```
address-family ipv6
  neighbor 2001:ACAD:B::1 activate
  neighbor 2001:ACAD:D::1 activate
  network 2001:B::/64
  network 2001:C::/64
  network 2001:ACAD:C::/64
exit-address-family
```

```
ip route 2.0.0.1 255.255.255.255 11.0.0.1
ip route 4.0.0.1 255.255.255.255 12.0.0.2
```

```
ipv6 route 2001:ACAD:B::/64 2001:B::1
ipv6 route 2001:ACAD:D::/64 2001:C::2
ipv6 router ospf 1
```

- **Show IP BGP**

Network	Next Hop	Metric	LocPrf
Weight Path			
*>i1.0.0.1/32	10.0.0.1	0	100
0 ?			
* i	10.0.0.1	0	100
0 ?			
r i2.0.0.1/32	11.0.0.1	0	100
0 ?			
r>i	11.0.0.1	0	100
0 ?			
r>i4.0.0.1/32	12.0.0.2	0	100
0 ?			
r i	12.0.0.2	0	100
0 ?			
* i5.0.0.1/32	13.0.0.2	0	100
0 ?			
*>i	13.0.0.2	0	100
0 ?			
r>i10.0.0.0/24	2.0.0.1	0	100
0 i			

```

r i          32.1.172.173          0    100
0 i
* i11.0.0.0/24    2.0.0.1          0    100
0 i
* i          32.1.172.173          0    100
0 i
*>          0.0.0.0          0
32768 i
* i12.0.0.0/24    32.1.172.173          0    100
0 i
* i          4.0.0.1          0    100
0 i
*>          0.0.0.0          0
32768 i
r i13.0.0.0/24    32.1.172.173          0    100
0 i
      Network      Next Hop      Metric LocPrf
Weight Path
r>i          4.0.0.1          0    100
0 i

```

- **Show IP BGP Summary**

```

BGP router identifier 3.3.3.3, local AS number 20
BGP table version is 16, main routing table version 16
8 network entries using 936 bytes of memory
18 path entries using 936 bytes of memory
4/3 BGP path/bestpath attribute entries using 496 bytes of
memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2368 total bytes of memory
BGP activity 20/3 prefixes, 38/7 paths, scan interval 60
secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
Up/Down	State/PfxRcd						
2.0.0.1	4	20	9927	9919	16	0	0
6d21h	4						
4.0.0.1	4	20	9926	9919	16	0	0
6d21h	4						

2001:ACAD:B::1	4	20	9931	9920	16	0	0
6d21h	4						
2001:ACAD:D::1	4	20	9930	9920	16	0	0
6d21h	4						

- **Show IP Route**

Gateway of last resort is not set

```

      1.0.0.0/32 is subnetted, 1 subnets
B       1.0.0.1 [200/0] via 10.0.0.1, 6d21h
      2.0.0.0/32 is subnetted, 1 subnets
S       2.0.0.1 [1/0] via 11.0.0.1
      3.0.0.0/32 is subnetted, 1 subnets
C       3.0.0.1 is directly connected, Loopback0
      4.0.0.0/32 is subnetted, 1 subnets
S       4.0.0.1 [1/0] via 12.0.0.2
      5.0.0.0/32 is subnetted, 1 subnets
B       5.0.0.1 [200/0] via 13.0.0.2, 6d21h
      10.0.0.0/24 is subnetted, 1 subnets
O E2    10.0.0.0 [110/1] via 11.0.0.1, 6d21h,
FastEthernet0/1
      11.0.0.0/24 is subnetted, 1 subnets
C       11.0.0.0 is directly connected, FastEthernet0/1
      12.0.0.0/24 is subnetted, 1 subnets
C       12.0.0.0 is directly connected, FastEthernet0/0
      13.0.0.0/24 is subnetted, 1 subnets
O E2    13.0.0.0 [110/1] via 12.0.0.2, 6d21h,
FastEthernet0/0

```

- **Show IPv6 Route**

```

B    2001:A::/64 [200/0]
     via 2001:ACAD:B::1
C    2001:B::/64 [0/0]
     via ::, FastEthernet0/1
L    2001:B::2/128 [0/0]
     via ::, FastEthernet0/1
C    2001:C::/64 [0/0]
     via ::, FastEthernet0/0
L    2001:C::1/128 [0/0]

```

```

        via ::, FastEthernet0/0
B  2001:D::/64 [200/0]
    via 2001:ACAD:D::1
B  2001:ACAD:A::/64 [200/0]
    via 2001:A::1
S  2001:ACAD:B::/64 [1/0]
    via 2001:B::1
C  2001:ACAD:C::/64 [0/0]
    via ::, Loopback0
L  2001:ACAD:C::1/128 [0/0]
    via ::, Loopback0
S  2001:ACAD:D::/64 [1/0]
    via 2001:C::2
B  2001:ACAD:E::/64 [200/0]
    via 2001:D::2
L  FE80::/10 [0/0]
    via ::, Null0
L  FF00::/8 [0/0]
    via ::, Null0

```

- **Ping R1**

- **F0/0**

- **10.0.0.1**

```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 36/54/64 ms

```

- **2001:a::1**

```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::1,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 44/53/64 ms

```

- **Lo0**

■ 1.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 40/57/96 ms

■ 2001:acad:a::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:A::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/54/72 ms

● Ping R2

○ F0/0

■ 11.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/29/36 ms

■ 2001:b::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/28/48 ms

○ F0/1

■ 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/29/40 ms

■ **2001:a::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/24/36 ms

○ **Lo0**

■ **2.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/24/40 ms

■ **2001:acad:b::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:B::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/27/40 ms

● **Ping R4**

○ **F0/0**

■ **13.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 13.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/31/36 ms

■ **2001:d::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 8/26/48 ms

○ **F0/1**

■ **12.0.0.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/30/48 ms

■ **2001:c::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 8/29/48 ms

○ **Lo0**

■ **4.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.0.0.1,
timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/34/48 ms

■ **2001:acad:d::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:ACAD:D::1,
timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/22/40 ms

● **Ping R5**

○ **F0/1**

■ **13.0.0.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 13.0.0.2,
timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/62/80 ms

■ **2001:d::2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:D::2,
timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/58/64 ms

○ **Lo0**

■ **5.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.0.0.1,
timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/52/68 ms

■ **2001:acad:e::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:E::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/50/64 ms

Router 4:

● **Show Run**

```
ipv6 unicast-routing

interface Loopback0
 ip address 4.0.0.1 255.255.255.255
 ipv6 address 2001:ACAD:D::1/64

interface FastEthernet0/0
 ip address 13.0.0.1 255.255.255.0
 ipv6 address 2001:D::1/64

interface FastEthernet0/1
 ip address 12.0.0.2 255.255.255.0
 ipv6 address 2001:C::2/64
 ipv6 ospf 1 area 0

router ospf 10
 router-id 4.4.4.4
 log-adjacency-changes
 redistribute bgp 20 subnets
 network 12.0.0.0 0.0.0.255 area 0

router bgp 20
 no synchronization
 bgp router-id 4.4.4.4
 bgp log-neighbor-changes
 network 4.0.0.0 mask 255.255.255.255
```

```
network 12.0.0.0 mask 255.255.255.0
network 13.0.0.0 mask 255.255.255.0
redistribute static
redistribute ospf 10
neighbor 2.0.0.1 remote-as 20
neighbor 2.0.0.1 update-source Loopback0
neighbor 3.0.0.1 remote-as 20
neighbor 3.0.0.1 update-source Loopback0
neighbor 5.0.0.1 remote-as 30
neighbor 5.0.0.1 ebgp-multihop 2
neighbor 5.0.0.1 update-source Loopback0
neighbor 2001:ACAD:B::1 remote-as 20
neighbor 2001:ACAD:B::1 update-source Loopback0
neighbor 2001:ACAD:C::1 remote-as 20
neighbor 2001:ACAD:C::1 update-source Loopback0
neighbor 2001:ACAD:E::1 remote-as 30
neighbor 2001:ACAD:E::1 ebgp-multihop 2
neighbor 2001:ACAD:E::1 update-source Loopback0
no auto-summary
```

```
address-family ipv6
  neighbor 2001:ACAD:B::1 activate
  neighbor 2001:ACAD:C::1 activate
  neighbor 2001:ACAD:E::1 activate
  network 2001:C::/64
  network 2001:D::/64
  network 2001:ACAD:D::/64
  redistribute static
  redistribute ospf 10
  no synchronization
exit-address-family
```

```
ip route 2.0.0.1 255.255.255.255 11.0.0.1
ip route 3.0.0.1 255.255.255.255 12.0.0.1
ip route 5.0.0.1 255.255.255.255 13.0.0.2
```

```
ipv6 route 2001:ACAD:B::/64 2001:B::1
ipv6 route 2001:ACAD:C::/64 2001:C::1
ipv6 route 2001:ACAD:E::/64 2001:D::2
ipv6 router ospf 1
```

● **Show IP BGP**

Network Weight Path	Next Hop	Metric	LocPrf
* i1.0.0.1/32 0 ?	10.0.0.1	0	100
*>i 0 ?	10.0.0.1	0	100
*> 2.0.0.1/32 32768 ?	0.0.0.0	0	
* i3.0.0.1/32 0 ?	11.0.0.2	0	100
* i 0 ?	11.0.0.2	0	100
*> 32768 ?	12.0.0.1	0	
*> 5.0.0.1/32 32768 ?	13.0.0.2	0	
r i10.0.0.0/24 0 i	32.1.172.173	0	100
r>i 0 i	2.0.0.1	0	100
* i11.0.0.0/24 0 i	32.1.172.173	0	100
* i 0 i	2.0.0.1	0	100
* i 0 i	32.1.172.173	0	100
* i 0 i	3.0.0.1	0	100
*> 32768 ?	12.0.0.1	2	
* i12.0.0.0/24 0 ?	11.0.0.2	2	100
* i 0 ?	11.0.0.2	2	100
* i 0 i	32.1.172.173	0	100
Network Weight Path	Next Hop	Metric	LocPrf

```

* i          3.0.0.1          0    100
0 i
*>          0.0.0.0          0
32768 i
* 13.0.0.0/24 32.1.172.173    0
0 30 i
*          5.0.0.1          0
0 30 i
*>          0.0.0.0          0
32768 i

```

- **Show IP BGP Summary**

```

BGP router identifier 4.4.4.4, local AS number 20
BGP table version is 12, main routing table version 12
8 network entries using 936 bytes of memory
22 path entries using 1144 bytes of memory
8/5 BGP path/bestpath attribute entries using 992 bytes of
memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3096 total bytes of memory
BGP activity 21/4 prefixes, 45/6 paths, scan interval 60
secs

```

Neighbor Up/Down	V State/PfxRcd	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
2.0.0.1 6d21h	4 5	20	9940	9940	12	0	0
3.0.0.1 6d21h	4 2	20	9933	9940	12	0	0
5.0.0.1 6d21h	4 1	30	9933	9939	12	0	0
2001:ACAD:B::1 6d21h	4 5	20	9944	9944	12	0	0
2001:ACAD:C::1 6d21h	4 2	20	9934	9944	12	0	0
2001:ACAD:E::1 6d21h	4 1	30	9934	9944	12	0	0

- **Show IP Route**

Gateway of last resort is not set

```

      1.0.0.0/32 is subnetted, 1 subnets
B       1.0.0.1 [200/0] via 10.0.0.1, 6d21h
      2.0.0.0/32 is subnetted, 1 subnets
S       2.0.0.1 [1/0] via 11.0.0.1
      3.0.0.0/32 is subnetted, 1 subnets
S       3.0.0.1 [1/0] via 12.0.0.1
      4.0.0.0/32 is subnetted, 1 subnets
C       4.0.0.1 is directly connected, Loopback0
      5.0.0.0/32 is subnetted, 1 subnets
S       5.0.0.1 [1/0] via 13.0.0.2
      10.0.0.0/24 is subnetted, 1 subnets
O E2    10.0.0.0 [110/1] via 12.0.0.1, 6d21h,
FastEthernet0/1
      11.0.0.0/24 is subnetted, 1 subnets
O       11.0.0.0 [110/2] via 12.0.0.1, 6d21h,
FastEthernet0/1
      12.0.0.0/24 is subnetted, 1 subnets
C       12.0.0.0 is directly connected, FastEthernet0/1
      13.0.0.0/24 is subnetted, 1 subnets
C       13.0.0.0 is directly connected, FastEthernet0/0
```

- **Show IPv6 Route**

```

B    2001:A::/64 [200/0]
      via 2001:ACAD:B::1
O    2001:B::/64 [110/2]
      via FE80::CE03:40FF:FE64:0, FastEthernet0/1
C    2001:C::/64 [0/0]
      via ::, FastEthernet0/1
L    2001:C::2/128 [0/0]
      via ::, FastEthernet0/1
C    2001:D::/64 [0/0]
      via ::, FastEthernet0/0
L    2001:D::1/128 [0/0]
      via ::, FastEthernet0/0
B    2001:ACAD:A::/64 [200/0]
      via 2001:A::1
```

```

S    2001:ACAD:B::/64 [1/0]
     via 2001:B::1
S    2001:ACAD:C::/64 [1/0]
     via 2001:C::1
C    2001:ACAD:D::/64 [0/0]
     via ::, Loopback0
L    2001:ACAD:D::1/128 [0/0]
     via ::, Loopback0
S    2001:ACAD:E::/64 [1/0]
     via 2001:D::2
L    FE80::/10 [0/0]
     via ::, Null0
L    FF00::/8 [0/0]
     via ::, Null0

```

- **Ping R1**

- **F0/0**

- **10.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.0.1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 64/86/108 ms

- **2001:a::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:A::1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 76/86/108 ms

- **Lo0**

- **1.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 1.0.0.1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 76/91/108 ms

■ **2001:acad:a::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:ACAD:A::1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 64/82/100 ms

● **Ping R2**

○ **F0/0**

■ **11.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 11.0.0.1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/59/76 ms

■ **2001:b::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:B::1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 20/44/64 ms

○ **F0/1**

■ **10.0.0.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.0.2,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 44/53/68 ms

■ **2001:a::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/56/80 ms

○ **Lo0**

■ **2.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 40/56/80 ms

■ **2001:acad:b::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:B::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/52/68 ms

● **Ping R3**

○ **F0/0**

■ **12.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1,
timeout is 2 seconds:
!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/31/40 ms

■ **2001:c::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/24/36 ms

○ **F0/1**

■ **11.0.0.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.0.0.2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/28/40 ms

■ **2001:b::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/27/44 ms

○ **Lo0**

■ **3.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 24/32/40 ms

■ **2001:acad:c::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:ACAD:C::1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/22/32 ms

● **Ping R5**

○ **F0/1**

■ **13.0.0.2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 13.0.0.2,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/29/48 ms

■ **2001:d::2**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:D::2,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/23/28 ms

○ **Lo0**

■ **5.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 5.0.0.1,
timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 20/30/36 ms

■ **2001:acad:e::1**

```
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 2001:ACAD:E::1,  
timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip  
min/avg/max = 16/27/36 ms
```

Router 5:

- **Show Run**

```
ipv6 unicast-routing  
  
interface Loopback0  
 ip address 5.0.0.1 255.255.255.255  
 ipv6 address 2001:ACAD:E::1/64  
  
interface FastEthernet0/1  
 ip address 13.0.0.2 255.255.255.0  
 ipv6 address 2001:D::2/64  
  
router bgp 30  
 no synchronization  
 bgp router-id 5.5.5.5  
 bgp log-neighbor-changes  
 network 5.0.0.0 mask 255.255.255.255  
 network 13.0.0.0 mask 255.255.255.0  
 neighbor 4.0.0.1 remote-as 20  
 neighbor 4.0.0.1 ebgp-multihop 2  
 neighbor 4.0.0.1 update-source Loopback0  
 neighbor 2001:ACAD:D::1 remote-as 20  
 neighbor 2001:ACAD:D::1 ebgp-multihop 2  
 neighbor 2001:ACAD:D::1 update-source Loopback0  
 no auto-summary  
  
address-family ipv6  
 neighbor 2001:ACAD:D::1 activate  
 network 2001:D::/64  
 network 2001:ACAD:E::/64  
 exit-address-family
```

```
ip route 4.0.0.1 255.255.255.255 13.0.0.1
```

```
ipv6 route 2001:ACAD:D::/64 2001:D::1
```

- **Show IP BGP**

Network	Next Hop	Metric	LocPrf
Weight Path			
*> 1.0.0.1/32	4.0.0.1		
0 20 ?			
*	32.1.172.173		
0 20 ?			
*> 2.0.0.1/32	4.0.0.1	0	
0 20 ?			
*	32.1.172.173	0	
0 20 ?			
*> 3.0.0.1/32	4.0.0.1	0	
0 20 ?			
*	32.1.172.173	0	
0 20 ?			
r> 5.0.0.1/32	4.0.0.1	0	
0 20 ?			
r	32.1.172.173	0	
0 20 ?			
*> 10.0.0.0/24	4.0.0.1		
0 20 i			
*	32.1.172.173		
0 20 i			
*> 11.0.0.0/24	4.0.0.1	2	
0 20 ?			
*	32.1.172.173	2	
0 20 ?			
*> 12.0.0.0/24	4.0.0.1	0	
0 20 i			
*	32.1.172.173	0	
0 20 i			
* 13.0.0.0/24	4.0.0.1	0	
0 20 i			
*	32.1.172.173	0	
0 20 i			

```
*>                                0.0.0.0                                0
32768 i
```

- **Show IP BGP Summary**

```
BGP router identifier 5.5.5.5, local AS number 30
BGP table version is 13, main routing table version 13
8 network entries using 936 bytes of memory
17 path entries using 884 bytes of memory
7/6 BGP path/bestpath attribute entries using 868 bytes of
memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2712 total bytes of memory
BGP activity 20/3 prefixes, 32/4 paths, scan interval 60
secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ
Up/Down	State/PfxRcd						
4.0.0.1	4	20	9957	9951	13	0	0
6d21h	8						
2001:ACAD:D::1	4	20	9962	9952	13	0	0
6d21h	8						

- **Show IP Route**

```
Gateway of last resort is not set
```

```

      1.0.0.0/32 is subnetted, 1 subnets
B       1.0.0.1 [20/0] via 4.0.0.1, 6d21h
      2.0.0.0/32 is subnetted, 1 subnets
B       2.0.0.1 [20/0] via 4.0.0.1, 6d21h
      3.0.0.0/32 is subnetted, 1 subnets
B       3.0.0.1 [20/0] via 4.0.0.1, 6d21h
      4.0.0.0/32 is subnetted, 1 subnets
S       4.0.0.1 [1/0] via 13.0.0.1
      5.0.0.0/32 is subnetted, 1 subnets
C       5.0.0.1 is directly connected, Loopback0
      10.0.0.0/24 is subnetted, 1 subnets
B       10.0.0.0 [20/0] via 4.0.0.1, 6d21h
```

```

    11.0.0.0/24 is subnetted, 1 subnets
B       11.0.0.0 [20/2] via 4.0.0.1, 6d21h
    12.0.0.0/24 is subnetted, 1 subnets
B       12.0.0.0 [20/0] via 4.0.0.1, 6d21h
    13.0.0.0/24 is subnetted, 1 subnets
C       13.0.0.0 is directly connected, FastEthernet0/1

```

- **Show IPv6 Route**

```

B    2001:A::/64 [20/0]
     via 2001:ACAD:D::1
B    2001:B::/64 [20/0]
     via 2001:ACAD:D::1
B    2001:C::/64 [20/0]
     via 2001:ACAD:D::1
C    2001:D::/64 [0/0]
     via ::, FastEthernet0/1
L    2001:D::2/128 [0/0]
     via ::, FastEthernet0/1
B    2001:ACAD:A::/64 [20/0]
     via 2001:ACAD:D::1
B    2001:ACAD:B::/64 [20/0]
     via 2001:ACAD:D::1
B    2001:ACAD:C::/64 [20/0]
     via 2001:ACAD:D::1
S    2001:ACAD:D::/64 [1/0]
     via 2001:D::1
C    2001:ACAD:E::/64 [0/0]
     via ::, Loopback0
L    2001:ACAD:E::1/128 [0/0]
     via ::, Loopback0
L    FE80::/10 [0/0]
     via ::, Null0
L    FF00::/8 [0/0]
     via ::, Null0

```

- **Ping R1**

- **F0/0**

- **10.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 96/137/200 ms

■ **2001:a::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 96/115/152 ms

○ **Lo0**

■ **1.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 108/132/164 ms

■ **2001:acad:a::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:A::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 92/120/168 ms

● **Ping R2**

○ **F0/0**

■ **11.0.0.1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 11.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 64/92/120 ms

■ **2001:b::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/84/120 ms

○ **F0/1**

■ **10.0.0.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 64/87/100 ms

■ **2001:a::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 44/82/120 ms

○ **Lo0**

■ **2.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.0.0.1,
timeout is 2 seconds:
!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 76/96/112 ms

■ **2001:acad:b::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:B::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 76/81/88 ms

● **Ping R3**

○ **F0/0**

■ **12.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/52/64 ms

■ **2001:c::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 40/57/76 ms

○ **F0/1**

■ **11.0.0.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.0.0.2,
timeout is 2 seconds:
!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/53/76 ms

■ **2001:b::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 32/52/92 ms

○ **Lo0**

■ **3.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/64/76 ms

■ **2001:acad:c::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:ACAD:C::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/48/64 ms

● **Ping R4**

○ **F0/0**

■ **13.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 13.0.0.1,
timeout is 2 seconds:
!!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 20/36/44 ms

■ **2001:d::1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/28/36 ms

○ **F0/1**

■ **12.0.0.2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/28/44 ms

■ **2001:c::2**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/25/36 ms

○ **Lo0**

■ **4.0.0.1**

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.0.0.1,
timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 20/32/44 ms

■ **2001:acad:d::1**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:ACAD:D::1,
timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/24/36 ms

Problems

One issue I had during this lab was that after I thought I was finished configuring the routers, I discovered I could ping all IPv4 addresses on all routers, but not the IPv6 addresses. After looking through my configurations thoroughly, I realized I had not activated the neighbor statements to allow BGP in IPv6 to be able to exchange information.

Another issue I had was that I could not ping all the loopback interfaces which I had set up in order to connect the middle BGP routers virtually. I tried researching how to get iBGP to work on your routers so that they could communicate but couldn't seem to find the answer I was looking for. Eventually, I talked with some of my peers, and one of them had the same issue as me. We discovered together that if we configured redistribute commands, specifically "redistribute connected", "redistribute static", and the "redistribute <process ID>" commands between BGP and OSPF, then we could ping all our loopback addresses from any router. However, another peer told me that the "redistribute connected" command is kind of a way to cheat and get everything to work and should only be used as a last resort. I did some testing and fiddling with the commands I had put in and found that if I got rid of the "redistribute connected" command, I was still able to ping all interfaces and loopbacks in both IPv4 and IPv6. Thus, I found I needed the redistribute commands between BGP and OSPF as well as the "redistribute static" command to allow everything to ping.

Conclusion

Ultimately from this lab, I learned how to set up iBGP so that it can communicate between other iBGP routers and to eBGP routers using both physical and virtual a.k.a. loopback interfaces. By challenging myself to do this in both IPv4 and IPv6 at the same time, I was able to directly see the difference and similarities in how iBGP is set up and how to connect it to eBGP. I learned more about the complexities of the BGP routing protocol and I was reminded of the difficulties one may have when tasked with writing a large number of commands and configuring routers in both IPv4 and IPv6. I am reminded to be careful and precise with commands as one small error when typing the command can cause various issues. I also was able to learn a few new commands and how they are important or contribute to the process. Overall, I was able to do well in this lab largely because I kept track of every command I wrote on each router and under what routing protocol or configuration mode they existed. Additionally, this lab gave me opportunities to communicate more with my peers to solve problems together. This lab proved to me that I can handle more complex network layouts.